

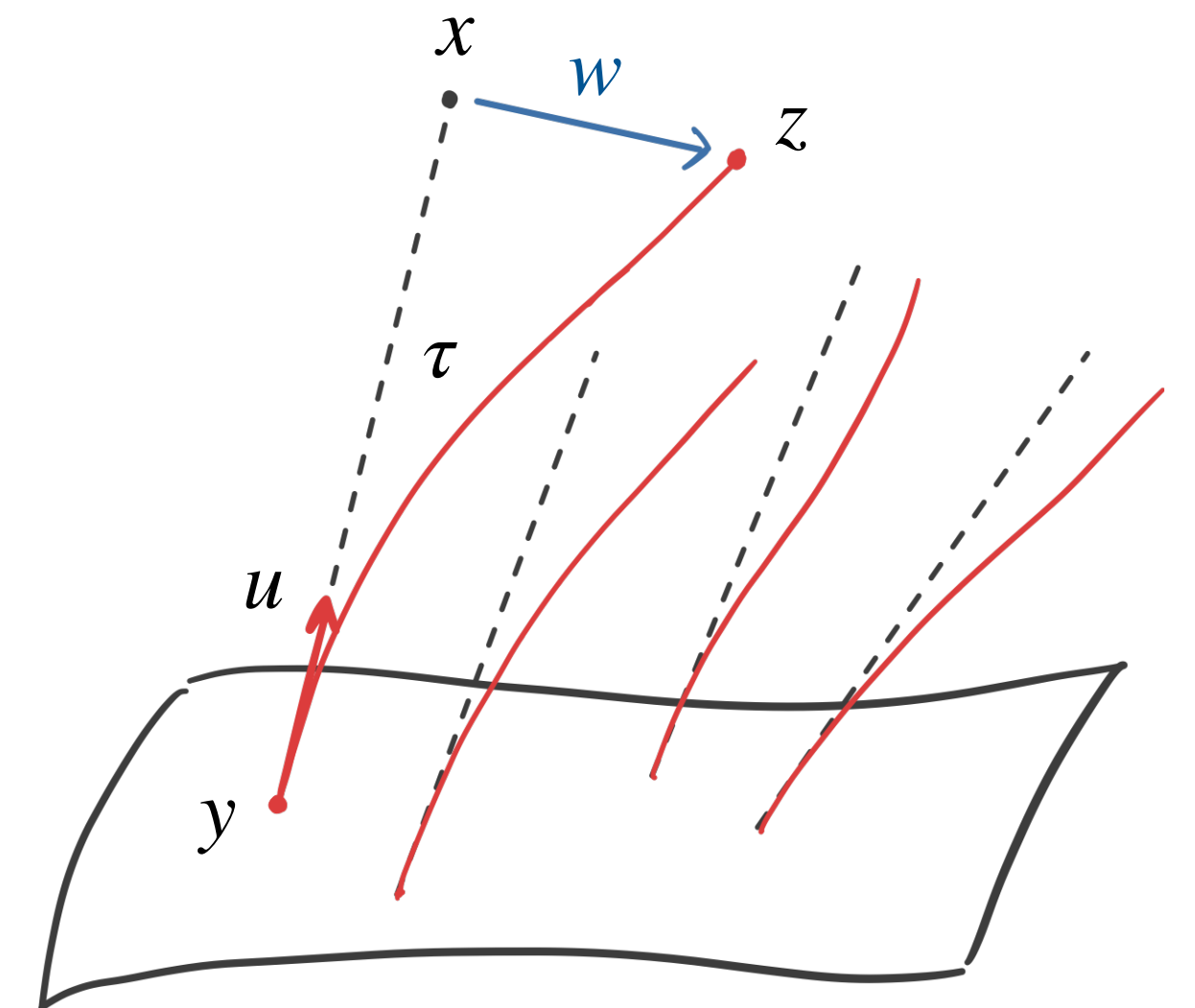
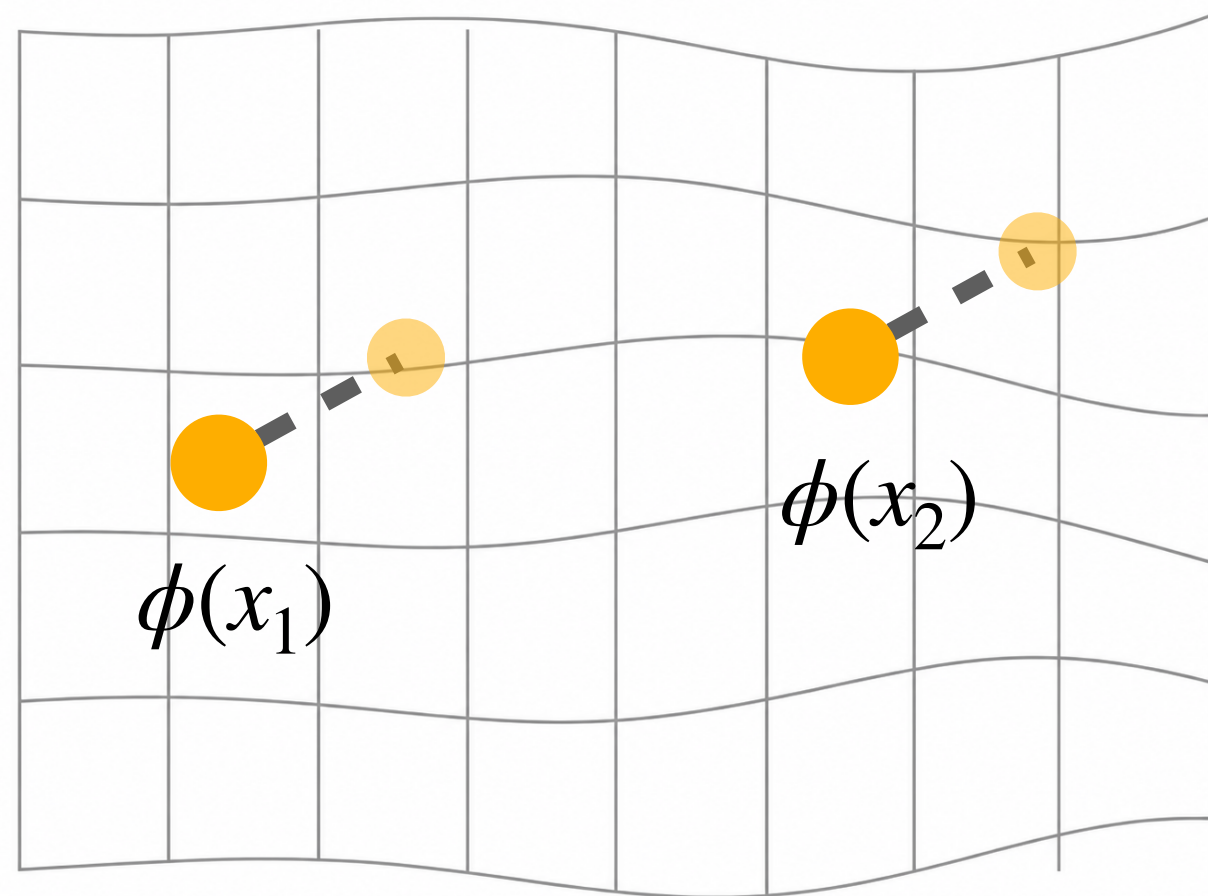
Calculus for Relational Observables

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Based on [2605.26077]

with Clifford Cheung, Allic Sivaramakrishnan
and Jordan Wilson-Gerow



How to define gauge-invariant bulk observables?

- QED:
- $U(1)$ gauge redundancy
 - Dressing by Wilson lines

$\psi(x)$



$$\Psi(x) = \psi(x) e^{i \int_\gamma u \cdot A}$$

[Dirac '55]

[Mandelstam '62]

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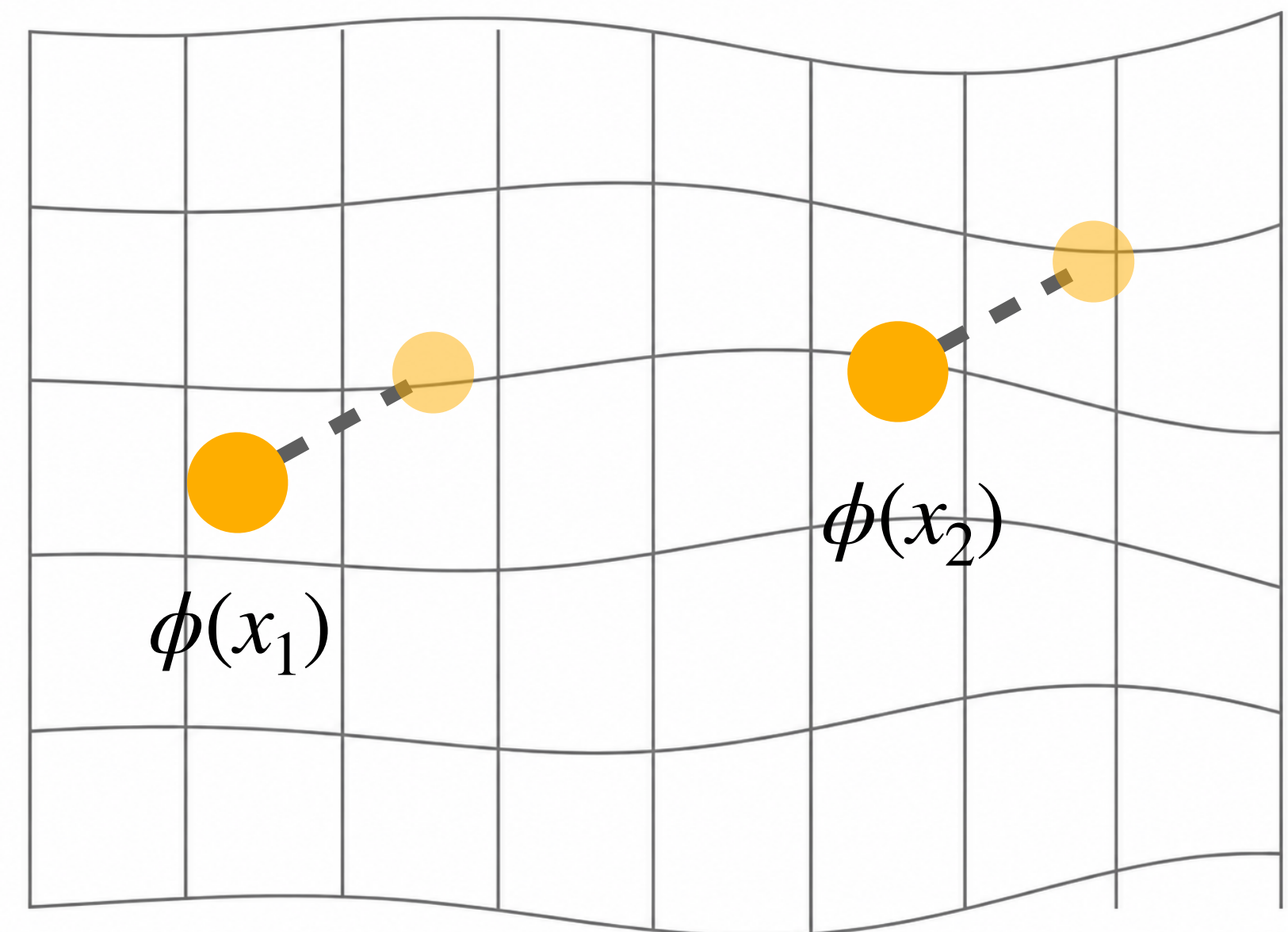
[Mandelstam '62]

Gravity:

[Donnelly & Giddings '15, '16]

- Gauge redundancy: coordinates

$$x \rightarrow x + \xi(x) \qquad \phi(x) \rightarrow \phi(x + \xi(x))$$



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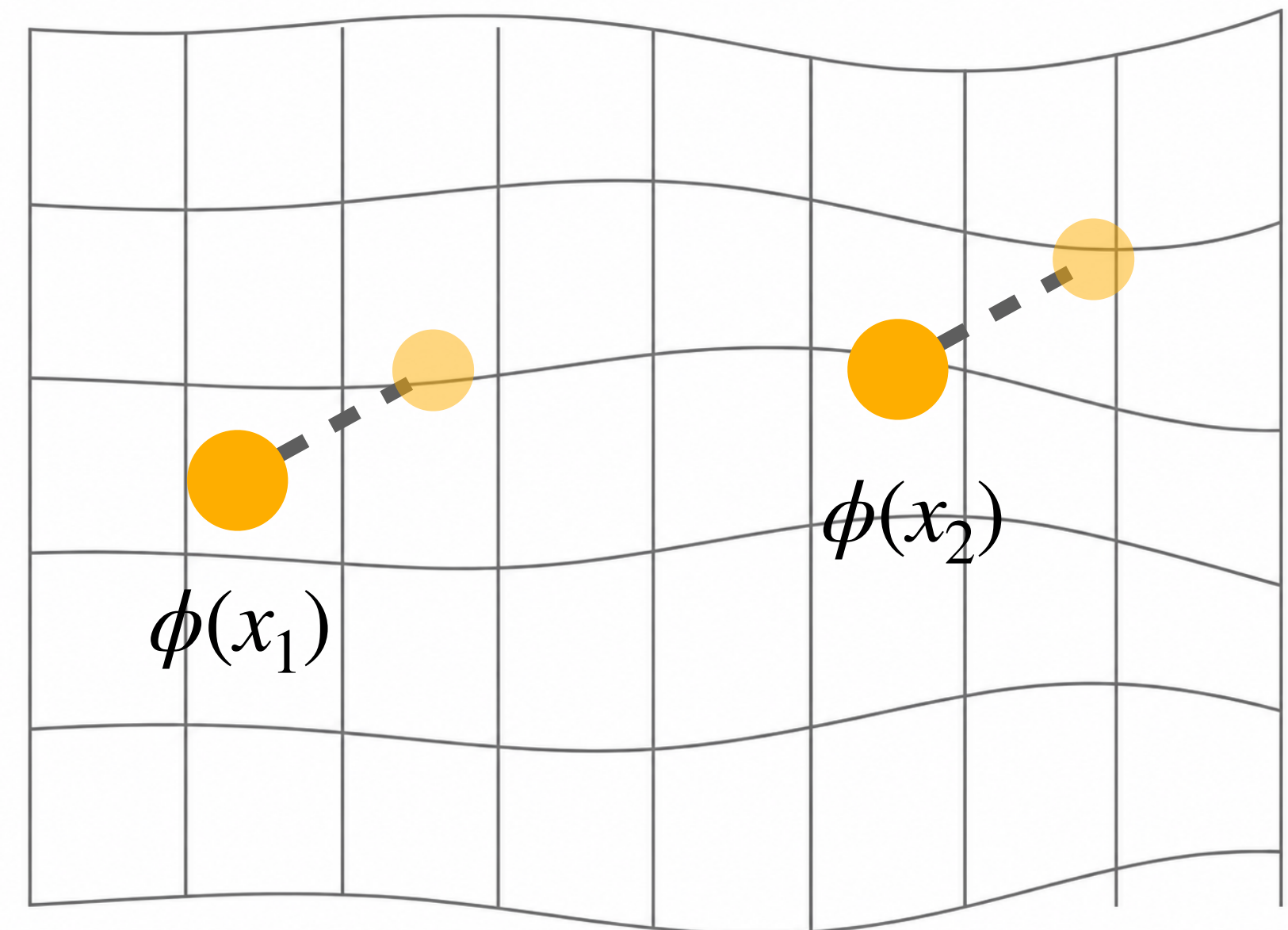
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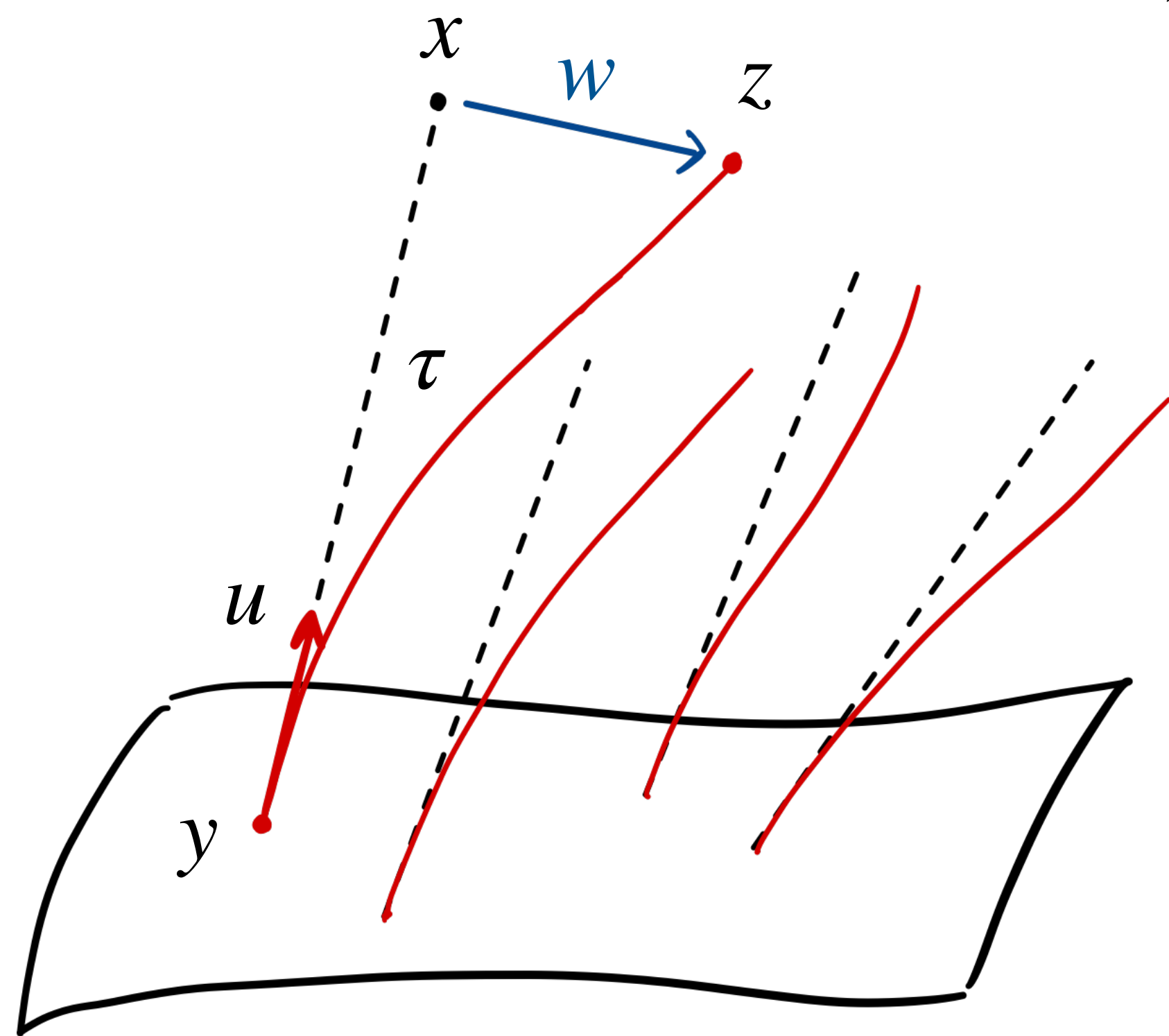
- Gauge redundancy: coordinates
 $x \rightarrow x + \xi(x)$ $\phi(x) \rightarrow \phi(x + \xi(x))$
- ~~$\langle 0 | \phi(x_1) \phi(x_2) | 0 \rangle$~~ : not gauge invariant
- “ x ” $\rightarrow$ should be defined *in relation to* physical reference system



Dressing

Geodesic dressing

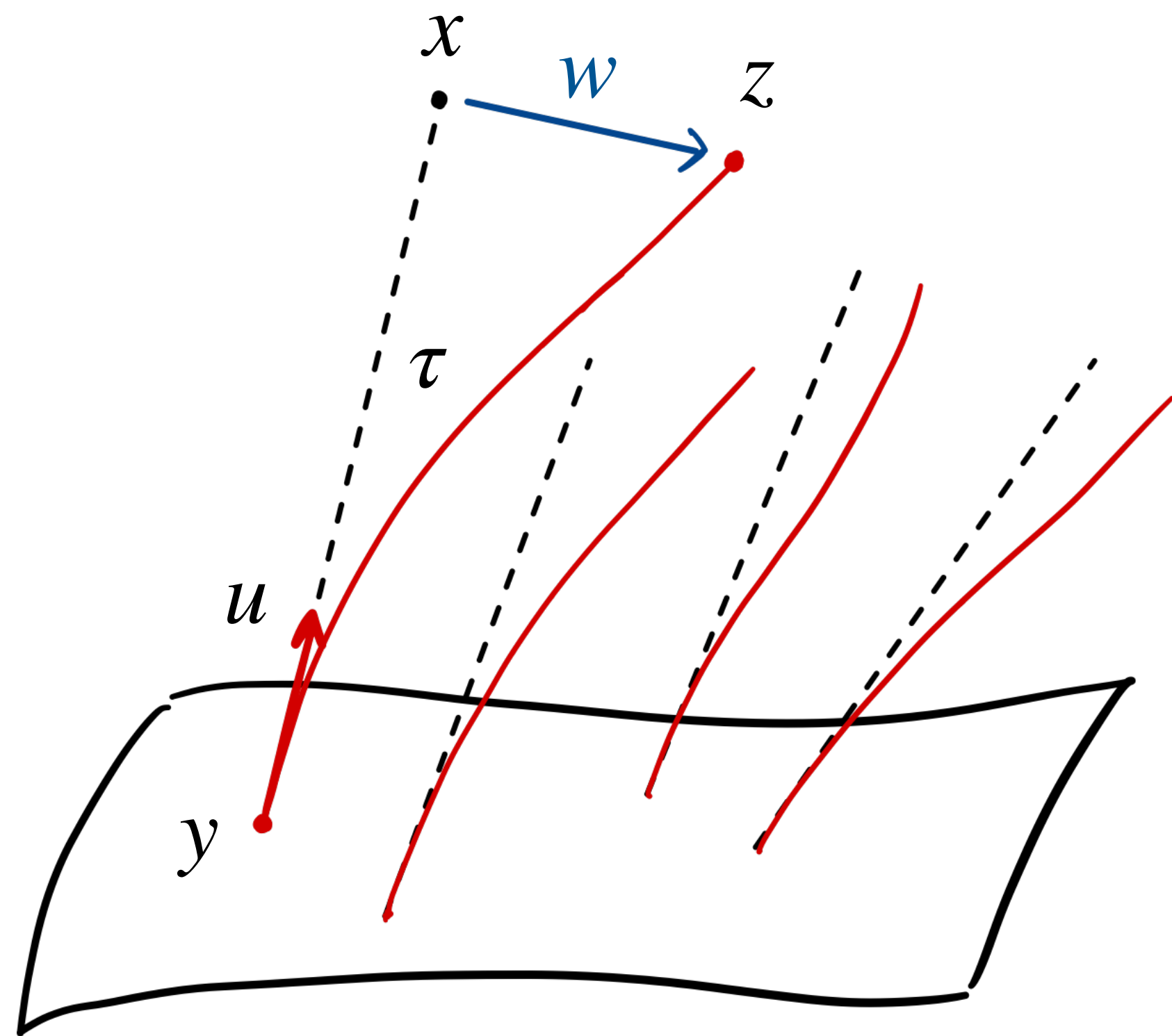
[Brown & Kuchar '94] [Donnelly & Giddings '16]



$$z^\mu(y, \tau) = \underbrace{y^\mu + u^\mu(y)(\tau - \tau_0)}_{\text{straight line } x^\mu} + \underbrace{w^\mu(y, \tau)}_{\text{deflection}}$$

Geodesic dressing

[Brown & Kuchar '94] [Donnelly & Giddings '16]



$$z^\mu(y, \tau) = \underbrace{y^\mu + u^\mu(y)(\tau - \tau_0)}_{\text{straight line } x^\mu} + \underbrace{w^\mu(y, \tau)}_{\text{deflection}}$$

- Dressing as a map:
flat-space coordinate $\rightarrow$ curved-space coordinate

$$x \rightarrow z(x)$$

- Dressed scalar $\Phi(x) \equiv \phi(z(x))$

$$\Phi(x) \equiv \phi(z(x))$$

Dressed scalar

[Donnelly & Giddings '16]

In first-order perturbation: $\Phi(x) = \phi(x) + w^\mu(x) \partial_\mu \phi(x) + \mathcal{O}(\kappa^2)$

where $w^\mu(x) = - \int_{-\infty}^0 d\tau \int_{-\infty}^s d\tau' \underbrace{\Gamma^\mu_{\alpha\beta}(x + u\tau') u^\alpha u^\beta}_{\propto h} + \mathcal{O}(\kappa^2)$

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- $\Phi(x)$:
- composite operator of ϕ and h
 - diff invariant

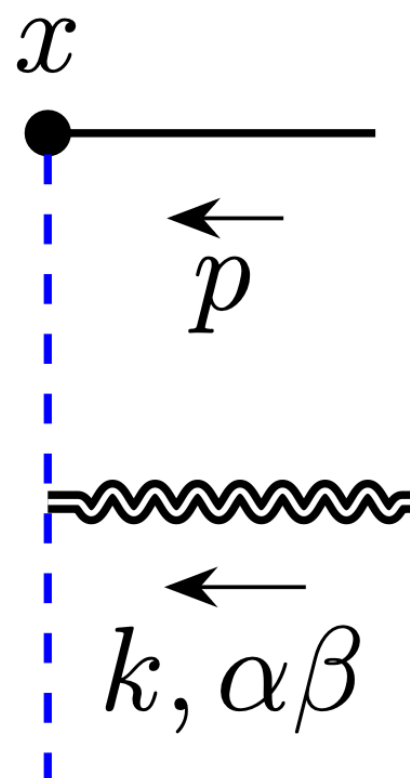
$$\delta_\xi \Gamma^\mu_{\alpha\beta} = \kappa \partial_\alpha \partial_\beta \xi^\mu \implies \delta_\xi w^\mu(x) = -\kappa \xi^\mu(x)$$

$$\begin{aligned} \delta_\xi \Phi(x) &= \delta_\xi \phi(x) + \delta_\xi w^\mu(x) \partial_\mu \phi(x) \\ &= \kappa \xi^\mu(x) \partial_\mu \phi(x) - \kappa \xi^\mu(x) \partial_\mu \phi(x) = 0 \end{aligned}$$

Feynman Rules

$$\Phi(x) = \phi(x) + w^\mu(x) \partial_\mu \phi(x) + \mathcal{O}(\kappa^2)$$

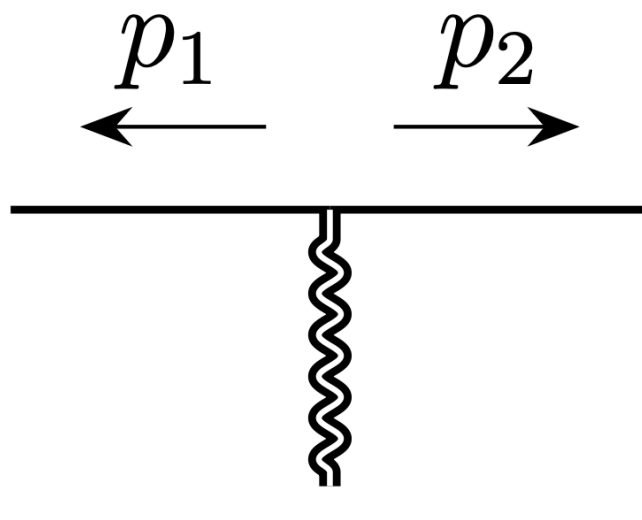
worldline dressing
vertex



The diagram shows a vertex at point x . A solid horizontal line extends to the right from x with a left-pointing arrow labeled p . A dashed vertical line extends downwards from x to point u . A wavy line connects x and u with a left-pointing arrow labeled $k, \alpha\beta$.

$$= \int_{p,k} e^{-ipx - ikx} \frac{-\kappa p^\mu \mathcal{O}_{\mu\alpha\beta}(k, u)}{(k \cdot u)^2}$$

scalar-scalar-graviton
3pt interaction

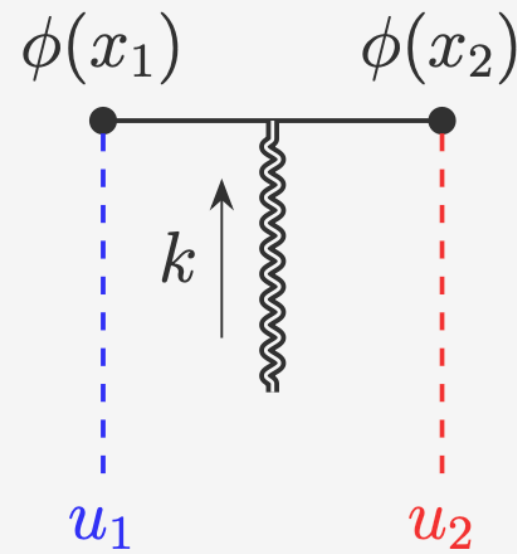


The diagram shows a horizontal solid line with two incoming arrows from the left labeled p_1 and p_2 . A wavy line extends downwards from the center of this line, labeled $\mu\nu$ at its end.

$$= i\kappa p_1^\alpha p_2^\beta P_{4\alpha\beta\mu\nu}$$

Tree-level Example: *gauge invariance*

undressed
correlator



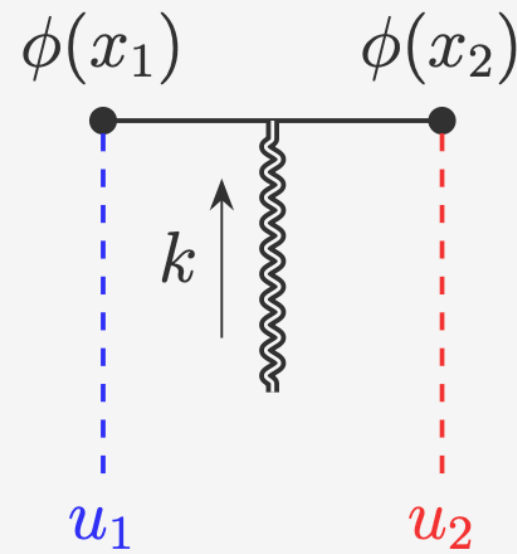
$$\langle 0 | \phi(x_1) \phi(x_2) | h(k) \rangle = \frac{(p_1 \cdot \epsilon \cdot p_2)}{p_1^2 p_2^2}$$

$\epsilon_{\mu\nu}$ shifts under gauge

$$\delta_\xi \epsilon_{\mu\nu} = -ik_\mu \tilde{\xi}_\nu - ik_\nu \tilde{\xi}_\mu$$

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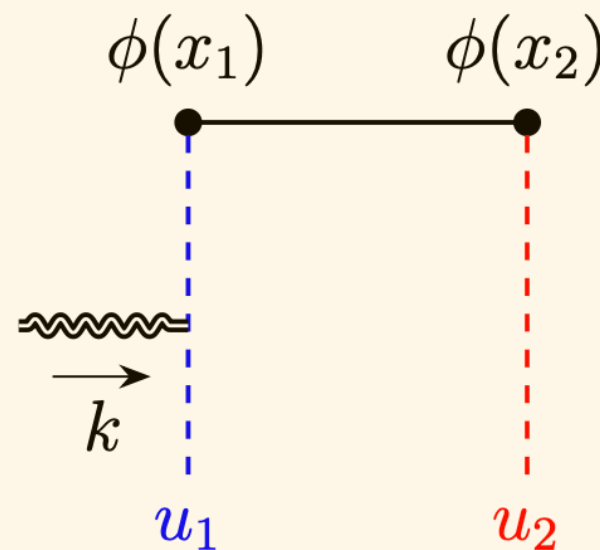


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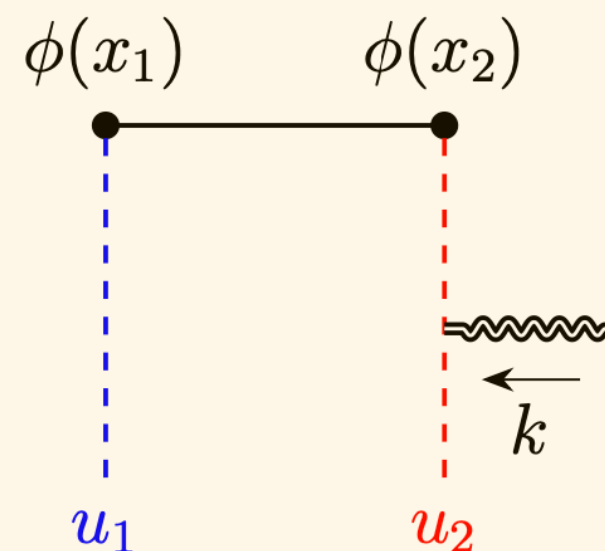
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dressing
corrections



$$\frac{1}{p_2^2} \frac{-2(p_2 \cdot \epsilon \cdot u_1)(k \cdot u_1) + (p_2 \cdot k)(u_1 \cdot \epsilon \cdot u_1)}{2(k \cdot u_1)^2}$$



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Sum of three diagrams:
gauge invariant!

$$\delta_\xi \left(\sum_{\text{diagrams}} \right) = 0$$

Tree-level Example: *kinematic singularities*

$$\langle 0 \left| \Phi(x_1) \Phi(x_2) \right| h(k) \rangle = -i\kappa \int_{p_1, p_2} e^{-ip_1 x_1 - ip_2 x_2} \times \delta(p_1 + p_2 - k) \\ \times \left[\frac{(p_1 \cdot \epsilon \cdot p_2)}{p_1^2 p_2^2} + \frac{1}{p_2^2} \frac{-2(p_2 \cdot \epsilon \cdot u_1)(k \cdot u_1) + (p_2 \cdot k)(u_1 \cdot \epsilon \cdot u_1)}{2(k \cdot u_1)^2} + \frac{1}{p_1^2} \frac{-2(p_1 \cdot \epsilon \cdot u_2)(k \cdot u_2) + (p_1 \cdot k)(u_2 \cdot \epsilon \cdot u_2)}{2(k \cdot u_2)^2} \right]$$

- Dressing is not an innocuous correction: $1/(k \cdot u_1)$ and $1/(k \cdot u_2)$ poles, which don't appear in usual correlation functions

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- Dressing is not an innocuous correction: $1/(k \cdot u_1)$ and $1/(k \cdot u_2)$ poles, which don't appear in usual correlation functions
- Detected if the external graviton is soft $k \rightarrow 0$, corresponding to the case where the worldline is on shell

Gauge Fixing

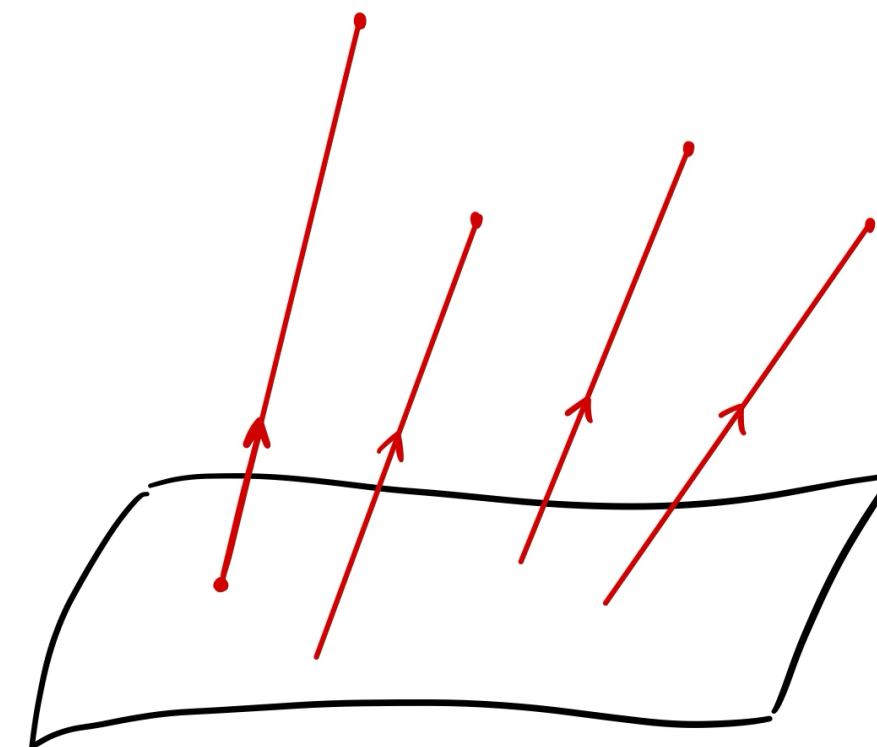
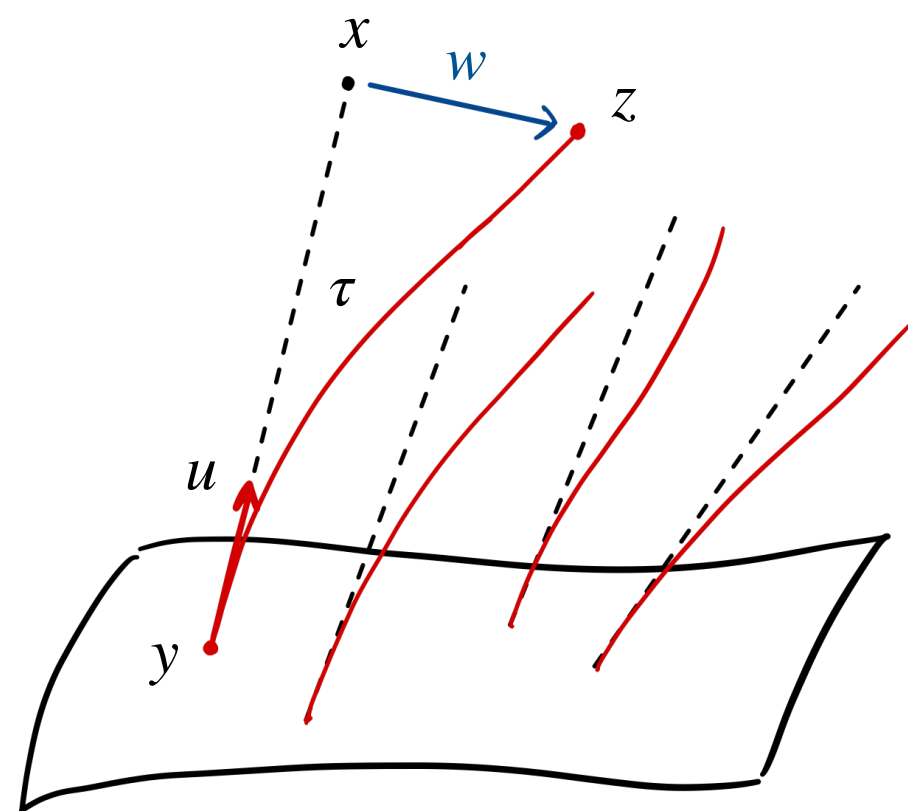
Dressing = gauge fixing

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calculations of dressed correlators don't depend on gauge

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- Since $\Phi(x) \equiv \phi(x + w(x))$ is gauge invariant, calculations of dressed correlators don't depend on gauge
- There's a convenient gauge s.t. $w^\mu(x) = 0$ everywhere, so dressing gets trivialized

$$\left\langle \prod_i \Phi_i \right\rangle_{\text{simple gauge}} = \left\langle \prod_i \phi_i \right\rangle_{\text{reference gauge}}$$



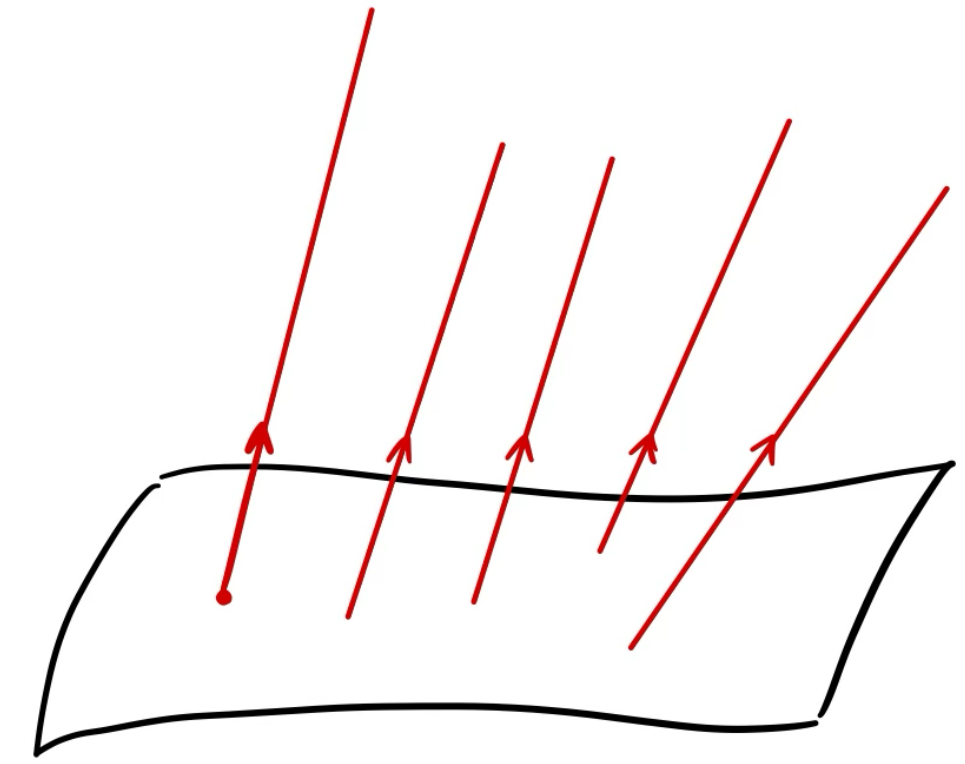
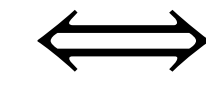
$$w^\mu(x) = 0$$

$$\Updownarrow$$

$$\Gamma^\mu_{\alpha\beta}(x) u^\alpha(x) u^\beta(x) = 0$$

Reference gauge:

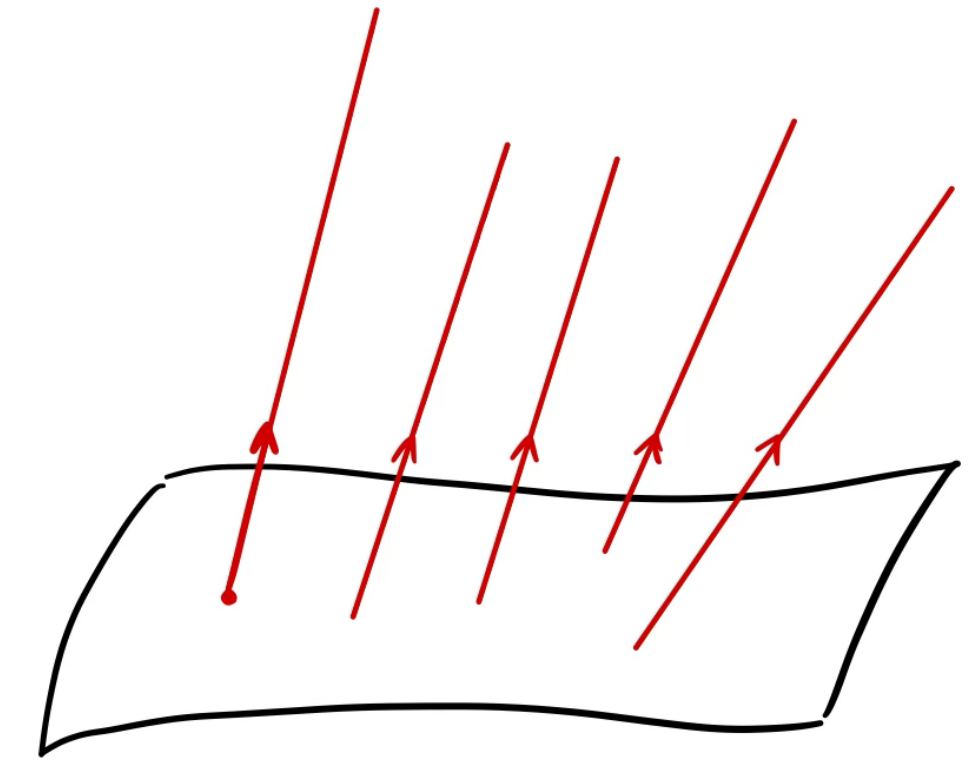
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$\Leftrightarrow$



Gauge transformation with $w^{\mu}(x)$ as the gauge parameter

$$\hat{\Gamma}^{\mu}_{\alpha\beta}(x) = \Gamma^{\mu}_{\alpha\beta}(x) + \partial_{\alpha} \partial_{\beta} w^{\mu}(x)$$

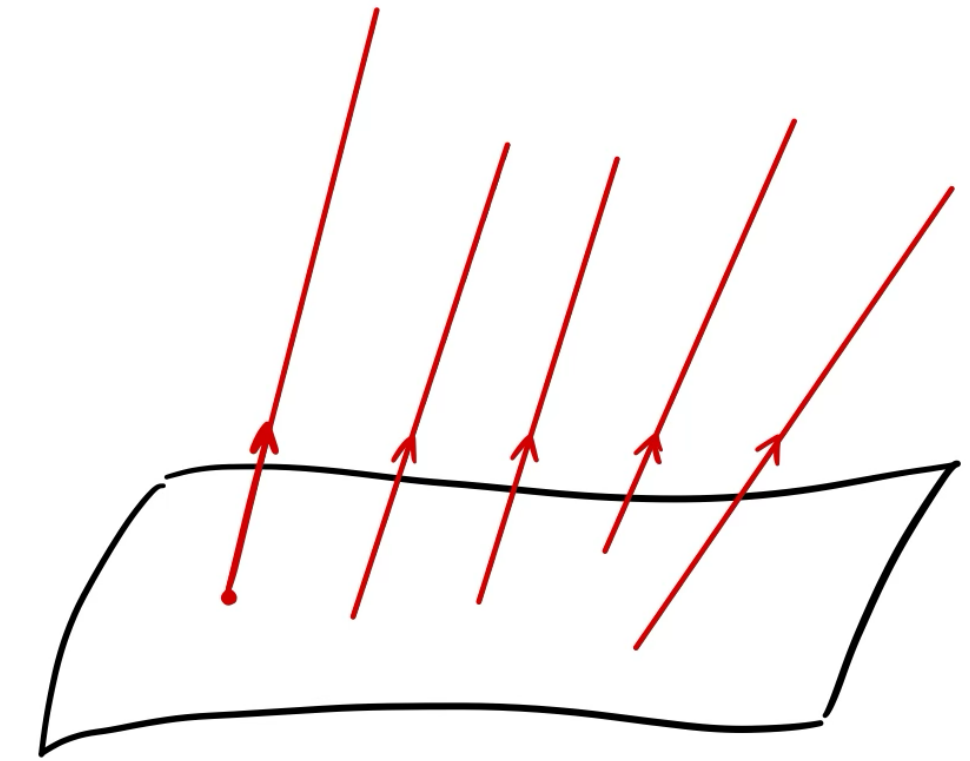
It satisfies

$$\begin{aligned} \hat{\Gamma}^{\mu}_{\alpha\beta}(x) u^{\alpha} u^{\beta} &= \Gamma^{\mu}_{\alpha\beta}(x) u^{\alpha} u^{\beta} + u^{\alpha} u^{\beta} \partial_{\alpha} \partial_{\beta} w^{\mu}(x) \\ &= \Gamma^{\mu}_{\alpha\beta}(x) u^{\alpha} u^{\beta} - \int_{-\infty}^0 d\tau \int_{-\infty}^{\tau} d\tau' \underbrace{u^{\alpha} u^{\beta} \partial_{\alpha} \partial_{\beta}}_{(u \cdot \partial)^2 = \partial_{\tau}^2} \left[\Gamma^{\mu}_{\rho\sigma}(x + u\tau') u^{\rho} u^{\sigma} \right] = 0 \end{aligned}$$

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The corresponding metric perturbation: $\hat{h}_{\mu\nu}(x) = h_{\mu\nu}(x) + \kappa^{-1} \partial_{\mu} w_{\nu}(x) + \kappa^{-1} \partial_{\nu} w_{\mu}(x)$

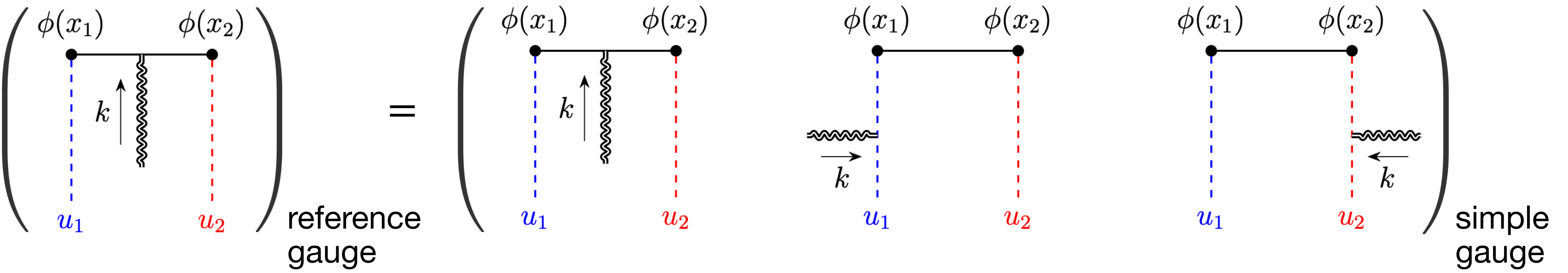
$$u^{\mu}(x) \hat{h}_{\mu\nu}(x) = 0$$

Gauge-fixed polarization

$$\langle 0 \left| \hat{h}_{\mu\nu}(x) \right| h(k) \rangle = \epsilon_{\mu\nu} e^{-ikx} - i\partial_\mu \left(\frac{\epsilon^{\alpha\beta} \mathcal{O}_{\nu\alpha\beta}(k, u(x))}{[k \cdot u(x)]^2} e^{-ikx} \right) - i\partial_\nu \left(\frac{\epsilon^{\alpha\beta} \mathcal{O}_{\mu\alpha\beta}(k, u(x))}{[k \cdot u(x)]^2} e^{-ikx} \right)$$

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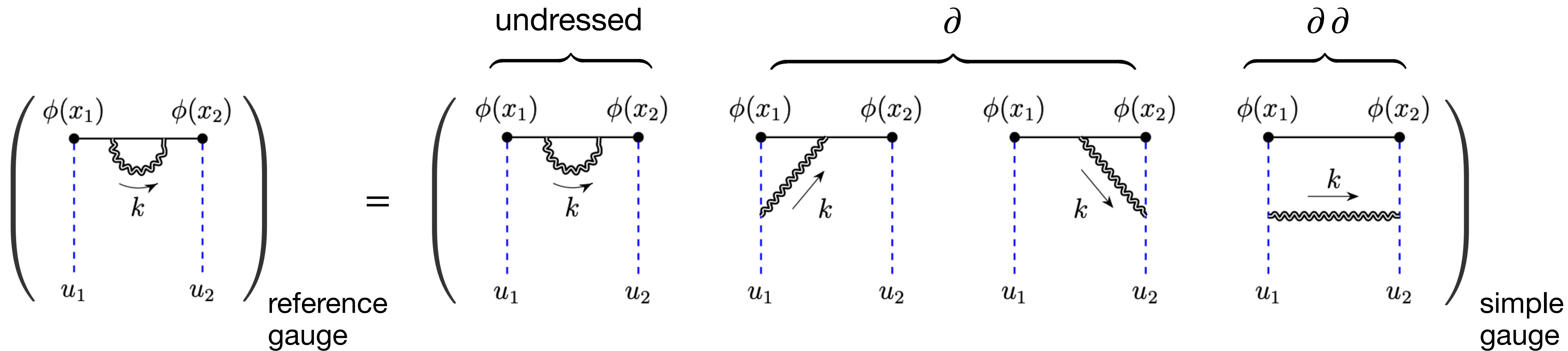


Gauge-fixed propagator

$$\begin{aligned} \langle 0 \left| \hat{h}_{\mu_1 \nu_1}(x_1) \hat{h}_{\mu_2 \nu_2}(x_2) \right| 0 \rangle &= \int_k \frac{i}{k^2} P_D^{\alpha_1 \beta_1 \alpha_2 \beta_2} \left[e^{-ikx_1} \eta_{\alpha_1 \mu_1} \eta_{\beta_1 \nu_1} - 2i \partial_{(\mu_1} \left(\frac{e^{-ikx_1} \mathcal{O}_{\nu_1) \alpha_1 \beta_1}(k, u(x_1))}{[k \cdot u(x_1)]^2} \right) \right] \\ &\quad \times \left[e^{ikx_2} \eta_{\alpha_2 \mu_2} \eta_{\beta_2 \nu_2} + 2i \partial_{(\mu_2} \left(\frac{e^{ikx_2} \mathcal{O}_{\nu_2) \alpha_2 \beta_2}(k, u(x_2))}{[-k \cdot u(x_2)]^2} \right) \right] \end{aligned}$$

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Summary

- Geolocate bulk points by dressing
Free-falling dust as dynamical clocks/rulers
- Dressing calculus
Worldline vertex, Feynman diagrams...
- Dressing is not ornamental
Kinematic poles $1/(k \cdot u)$ detected by external soft gravitons
- Dressing = gauge fixing
Reference gauge $u^\mu(x) h_{\mu\nu}(x) = 0$ with a spacetime-variant $u(x)$

Interesting calculations:

1. ADM mass from dressed correlators?
2. Hawking radiation?
3. Asymptotic symmetries?

Thanks!

$$\begin{array}{c} \overleftarrow{p_1} \qquad \overrightarrow{p_2} \\ \hline \text{wavy line} \\ \mu\nu \end{array} = i\kappa p_1^\alpha p_2^\beta P_{4\;\alpha\beta\mu\nu}$$

$$\begin{array}{c} x \\ \bullet \\ \text{---} \\ \overleftarrow{p} \\ \text{wavy line} \\ \overleftarrow{k, \alpha\beta} \\ \text{---} \\ u \end{array} = \int_{p,k} e^{-ipx-ikx} \frac{-\kappa p^\mu \mathcal{O}_{\mu\alpha\beta}(k,u)}{(k\cdot u)^2}$$

$$P_D^{\alpha\beta\gamma\delta} = \frac{1}{2} \Big(\eta^{\alpha\gamma} \eta^{\beta\delta} + \eta^{\alpha\delta} \eta^{\beta\gamma} - \frac{2}{D-2} \eta^{\alpha\beta} \eta^{\gamma\delta} \Big)$$

$$\mathcal{O}^{\mu\alpha\beta}(k,u(x)) = \frac{1}{2} \Big([k\cdot u(x)] u^\alpha(x) \eta^{\beta\mu} + [k\cdot u(x)] u^\beta(x) \eta^{\alpha\mu} - k^\mu u^\alpha(x) u^\beta(x) \Big)$$

$$w^\mu(x) = -\, i\kappa \int_k e^{-ikx} \frac{k_\beta h^\mu_\alpha(k) + k_\alpha h^\mu_\beta(k) - k^\mu h_{\alpha\beta}(k)}{2[k\cdot u(x)]^2} u^\alpha(x) u^\beta(x)$$